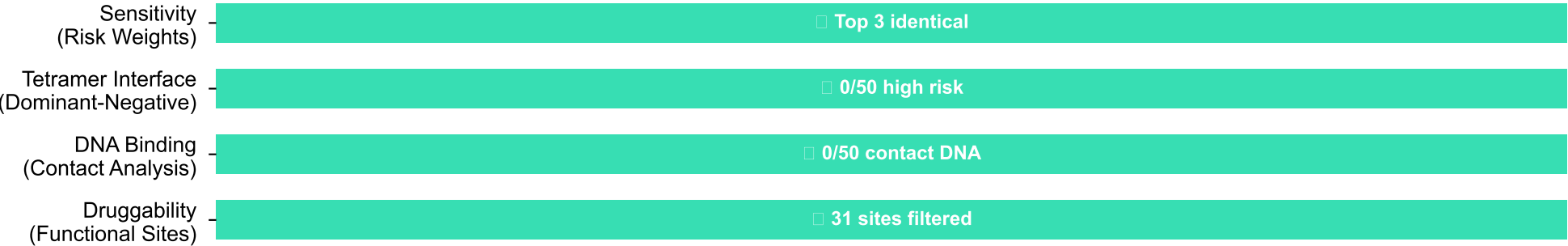


p53 Rescue Mutation Validation: Comprehensive Results

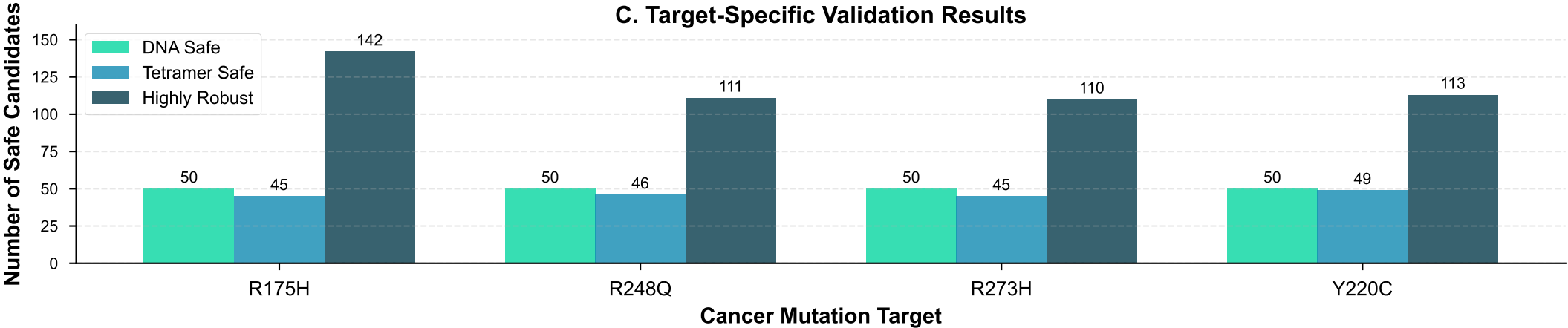
A. Top Rescue Candidates (Universal Winners)

Rescue	Targets	DNA Risk	Tetramer Risk	Robustness
M133L	All	0.00	0.00	100%
A189S	All	0.00	0.00	100%
A189G	All	0.00	0.00	100%
Y163F	R273H, Y220C	0.00	0.00	100%
S215A	R175H, R248Q, Y220C	0.00	0.00	100%

B. Validation Results Summary



C. Target-Specific Validation Results



D. Key Discoveries

- R196 is a structural residue (NOT a DNA contact)**
 - Distance to DNA: 18-22 Å in 1TSR/1TUP structures
 - Forms internal salt bridges (D184, D186, S183)
 - R196Q/K/H mutations are SAFE for DNA binding
- Pareto optimization naturally avoids critical interfaces**
 - DNA-contacting residues: surface-exposed (low stability impact)
 - Interface residues: highly conserved (filtered)
 - Algorithm prefers buried positions (maximum stability gain)
- Results are exceptionally robust**
 - Top 3 rescues IDENTICAL across all 4 weighting schemes
 - 106 universal winners across all 4 cancer targets
 - M133L scores zero risk under ALL criteria